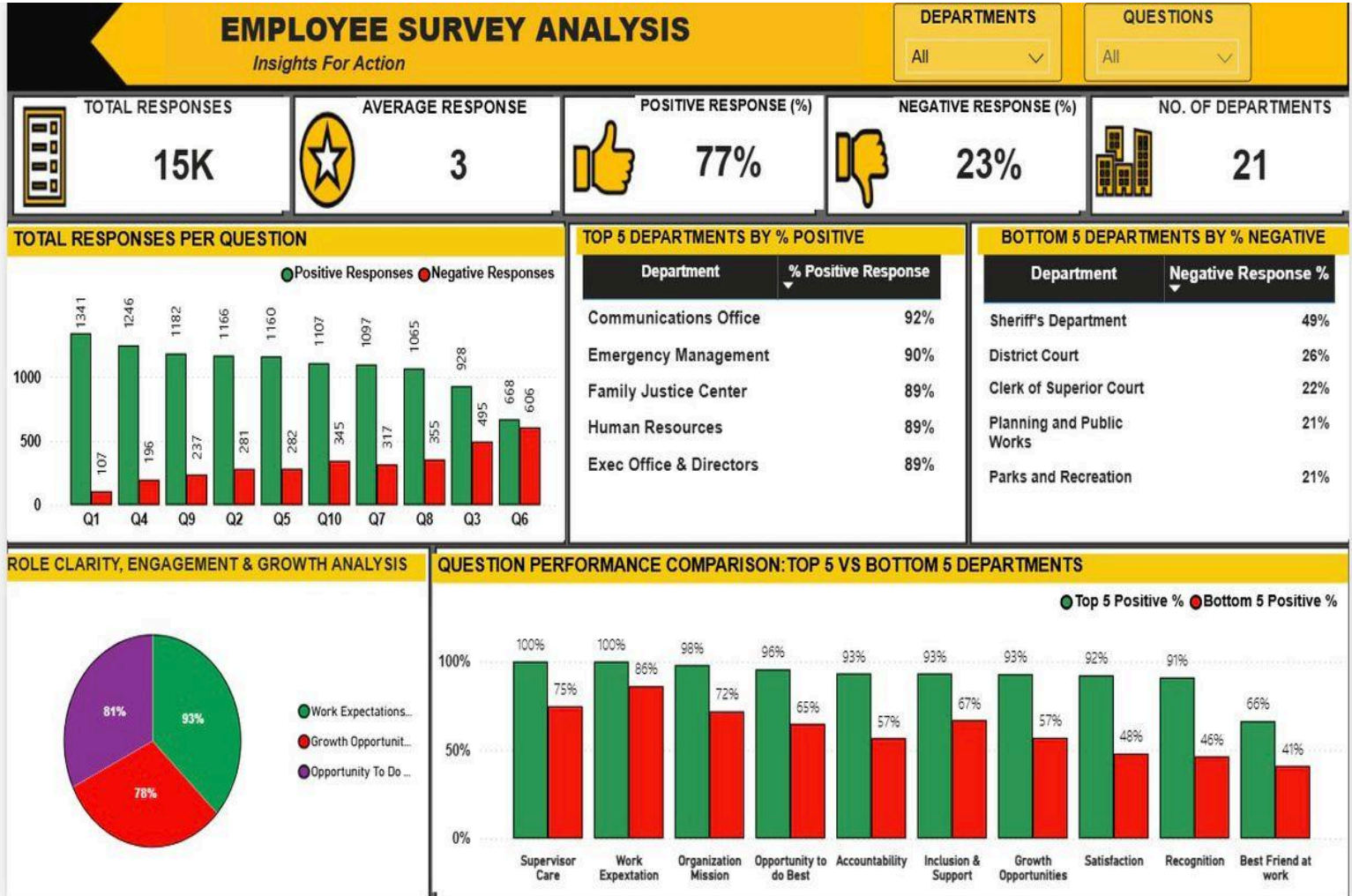


EMPLOYEE SURVEY DASHBOARD



EMPLOYEE SURVEY ANALYSIS REPORT

Executive Summary

This project analyzes employee engagement survey responses collected across 21 departments within Pierce County, Washington. The objective was to assess employee perceptions regarding work expectations, leadership, recognition, accountability, growth opportunities, inclusion, and overall job satisfaction.

Microsoft Power BI was used to clean, transform, model, and analyze the dataset. An interactive dashboard was developed to present key performance indicators (KPIs), departmental comparisons, and employee engagement trends.

The analysis revealed an overall positive employee sentiment, with a **77% positive response rate** compared to **23% negative responses**. While employees generally expressed positive perceptions regarding work expectations, supervisor support, and organizational mission, lower agreement levels were observed for recognition, growth opportunities, workplace relationships, and overall job satisfaction.

The dashboard enables management to monitor employee engagement, compare departmental performance, identify improvement areas, and make informed decisions based on employee feedback.

Introduction

Employee engagement surveys provide organizations with valuable insights into employees' perceptions of their workplace. Understanding employee experiences helps management identify organizational strengths, address workplace challenges, and improve employee satisfaction and performance.

This project analyzes employee survey responses collected across 21 departments within Pierce County, Washington. Using Microsoft Power BI, the dataset was cleaned, transformed, modeled, and visualized to evaluate employee engagement across ten survey questions. The project also compares departmental performance and identifies key factors influencing employee satisfaction.

Dataset Overview

The dataset contains employee survey responses collected from employees across 21 departments within Pierce County, Washington.

Each record represents an employee's response to one survey question and includes information relating to department, position, survey question, response value, and response description.

The survey consists of ten employee engagement questions covering:

- Work Expectations
- Opportunity to Do Best Every Day
- Recognition and Praise
- Supervisor Support
- Organizational Mission
- Best Friend at Work
- Growth Opportunities
- Accountability
- Inclusion and Workforce Support
- Overall Job Satisfaction

Survey responses were recorded using the following scale:

Response	Meaning
0	Not Applicable
1	Strongly Disagree
2	Disagree
3	Agree
4	Strongly Agree

For this analysis:

- Responses **3 (Agree)** and **4 (Strongly Agree)** were classified as **Positive**.
- Responses **1 (Strongly Disagree)** and **2 (Disagree)** were classified as **Negative**.
- Responses **0 (Not Applicable)** were excluded from sentiment calculations.

Data Cleaning and Preparation

Data cleaning and preparation were carried out using Power Query in Microsoft Power BI to improve data quality, ensure consistency, and prepare the dataset for analysis.

The following cleaning and transformation steps were performed:

Data Validation

- Reviewed the dataset for missing values, duplicate records, and inconsistencies.
- Verified that all survey responses contained valid department, question, and response information.
- Confirmed that Response IDs were unique and represented individual survey responses.
- The dataset was examined for duplicate records. Response identifiers were reviewed to ensure that each survey response represented a unique observation.

Data Type Formatting

The appropriate data types were assigned to each column to support accurate calculations and relationships.

Examples include:

- Response ID – Whole Number
- Department – Text
- Position – Text
- Question – Text
- Response – Whole Number
- Response Text – Text

Response Classification

Survey responses were grouped into sentiment categories for analysis.

- **Positive Responses:** Agree (3) and Strongly Agree (4)
- **Negative Responses:** Disagree (2) and Strongly Disagree (1)
- **Not Applicable (0):** Excluded from sentiment calculations

This classification formed the basis for calculating positive and negative response rates.

3.4 Question Standardization

To improve dashboard readability, shortened question names were created for the survey questions while retaining the original questions within the dimension table.

Examples include:

Original Question

I know what is expected of me at work

Short Name

Work Expectations

At work, I have the opportunity to do what I do best every day	Opportunity to Do Best
In the last seven days, I have received recognition or praise for doing good work	Recognition
Overall I am satisfied with my job	Job Satisfaction

The shortened names were used in dashboard visualizations to improve readability.

Data Preparation for Analysis

To support efficient reporting and analysis:

- Separate dimension tables were created for survey questions and employee positions.
- Response categories were standardized across the dataset.
- The dataset was prepared for star schema modeling in Power BI.

The cleaned dataset provided a consistent and reliable foundation for developing analytical measures and dashboard visualizations.

Data Modelling

A star schema data model was developed in Microsoft Power BI to improve data organization, reporting efficiency, and analytical performance.

The model consists of one fact table and two dimension tables.

Fact_Employee Survey Table

- **Fact_Employee Survey** – Stores individual employee survey responses, including department, position, question, response value, and Response ID.

Dimension Tables

- **Dim Question & Response** – Contains survey questions, shortened question names, and response descriptions.
- **Dim Position** – Contains employee position information used for filtering and analysis.

Relationships were established between the fact table and the dimension tables using appropriate key fields, enabling efficient filtering and aggregation across the dashboard.

The star schema simplified report development, reduced data redundancy, and improved the performance of DAX calculations and visualizations.

Measures and Calculations

Several DAX measures were created to support the analysis and dashboard development.

Measure	Description
Total Survey Responses	Counts the total number of survey responses.
Average Response Score	Calculates the average response score across all survey responses.
Positive Responses	Counts responses with values of 3 (Agree) and 4 (Strongly Agree).
Negative Responses	Counts responses with values of 1 (Strongly Disagree) and 2 (Disagree).
Positive Response Rate (%)	Calculates the percentage of positive responses, excluding Not Applicable responses.
Negative Response Rate (%)	Calculates the percentage of negative responses, excluding Not Applicable responses.
Number of Departments	Counts the total number of departments included in the survey.
Department Positive Response Rate	Calculates the positive response percentage for each department.

Department Negative Response Rate	Calculates the negative response percentage for each department.
Department Ranking	Ranks departments based on their positive response rate.
Best 5 Department Positive Rate	Calculates the average positive response rate for the top five departments.
Bottom 5 Department Positive Rate	Calculates the average positive response rate for the bottom five departments.

These measures formed the basis for calculating employee sentiment, departmental performance, and comparative analyses throughout the dashboard.

Dashboard Overview

An interactive Employee Survey Dashboard was developed in Microsoft Power BI to provide a comprehensive overview of employee engagement across the organization. The dashboard consolidates key performance indicators, departmental performance, and survey question responses into a single interface, enabling users to quickly assess employee sentiment and identify areas requiring management attention.

The dashboard includes executive summary KPIs, question level analysis, departmental comparisons, and interactive filters that allow users to explore employee responses by department and survey question. Its user friendly layout enables management to monitor organizational performance and make data driven decisions based on employee feedback.

Dashboard Objectives

The dashboard was designed to achieve the following objectives:

- Provide an overview of overall employee engagement.
- Measure positive and negative employee sentiment.
- Compare employee responses across all survey questions.
- Identify the highest and lowest performing departments.
- Compare employee experiences between top and bottom performing departments.
- Support management in identifying areas requiring improvement.

- Enable interactive exploration of survey results using dashboard filters.

Key Performance Indicators (KPIs)

The dashboard includes the following key performance indicators:

- Total Survey Responses
- Average Response Score
- Positive Response Rate (%)
- Negative Response Rate (%)
- Number of Departments
- Question Level Positive vs Negative Responses
- Top 5 Departments by Positive Response Rate (%)
- Bottom 5 Departments by Negative Response Rate (%)
- Question Performance Comparison: Top 5 vs Bottom 5 Departments
- Role Clarity, Engagement and Growth Analysis

Dashboard Visualizations

Executive KPI Cards

The dashboard presents five KPI cards summarizing the overall survey results, including the total number of survey responses, average response score, positive response rate, negative response rate, and total number of departments. These KPIs provide management with an immediate overview of employee sentiment across the organization.

Question Level Positive vs Negative Responses

A clustered column chart compares the number of positive and negative responses for each of the ten survey questions. This visualization highlights areas where employees expressed stronger agreement as well as questions receiving relatively higher negative responses.

Top 5 Departments by Positive Response Rate

A horizontal bar chart ranks the five departments with the highest positive response rates. This visualization identifies departments demonstrating the strongest employee engagement and satisfaction and provides useful benchmarks for organizational best practices.

Bottom 5 Departments by Negative Response Rate

A horizontal bar chart displays the five departments with the highest negative response rates. This visualization enables management to quickly identify departments requiring additional support and targeted improvement initiatives.

Role Clarity, Engagement and Growth Analysis

This visualization compares positive response rates for three key employee experience indicators: **Work Expectations**, **Opportunity to Do Best**, and **Growth Opportunities**. The comparison provides insight into employees' understanding of their roles, their ability to utilize their strengths, and their perceptions of professional development opportunities.

Question Performance Comparison: Top 5 vs Bottom 5 Departments

A clustered column chart compares the positive response percentages for each survey question between the top five and bottom five performing departments. This visualization highlights the employee experience factors that contribute most significantly to departmental performance differences and identifies key areas for organizational improvement.

Dashboard Analysis and Findings

The dashboard provides a comprehensive overview of employee engagement across the organization, highlighting overall sentiment, departmental performance, and employee perceptions across the ten survey questions.

Overall Employee Sentiment

The survey recorded approximately **15,000 responses** across **21 departments**. The overall analysis indicates a positive employee experience, with a **77% positive response rate** compared to a **23% negative response rate**. The average response score of **3** further suggests that employees generally agreed with the survey statements.

These results indicate that employees have a favorable perception of their workplace; however, opportunities remain to improve specific aspects of employee engagement.

Question Level Analysis

The comparison of positive and negative responses across all survey questions revealed noticeable differences in employee perceptions.

Questions relating to **Work Expectations**, **Supervisor Support**, **Mission and Purpose**, and **Inclusion and Support** received the highest levels of positive responses, indicating that employees generally understand their responsibilities, feel supported by their supervisors, and recognize the importance of their work within the organization.

Conversely, **Recognition**, **Best Friend at Work**, **Growth Opportunities**, and **Job Satisfaction** recorded comparatively higher levels of negative responses, suggesting that these areas require greater management attention.

Department Performance

Departmental analysis showed considerable variation in employee engagement across the organization.

The **Communications Office** recorded the highest positive response rate (**92%**), followed by **Emergency Management (90%)**, **Family Justice Center (89%)**, **Human Resources (89%)**, and **Executive Office & Directors (89%)**. These departments consistently demonstrated strong employee sentiment and may serve as benchmarks for organizational best practices.

In contrast, the **Sheriff's Department** recorded the highest negative response rate (**49%**), followed by the **District Court (26%)**, **Clerk of Superior Court (22%)**, **Planning and Public Works (21%)**, and **Parks and Recreation (21%)**. These findings suggest opportunities for targeted interventions to improve employee engagement within these departments.

Role Clarity, Engagement and Growth Analysis

Positive response rates were compared across three key employee experience indicators:

- Work Expectations (**93%**)
- Opportunity to Do Best (**81%**)
- Growth Opportunities (**78%**)

The results indicate that employees clearly understand what is expected of them and generally feel they have opportunities to utilize their strengths. However, positive responses decline when employees evaluate opportunities for learning and professional growth.

This suggests that while employees have clarity regarding their roles, additional focus is needed to strengthen career development and continuous learning opportunities.

Best vs Bottom Performing Departments

The comparison between the top five and bottom five departments provides valuable insight into the factors driving departmental performance.

Across nearly all survey questions, the top performing departments consistently achieved higher positive response rates than the bottom performing departments. The largest performance gaps were observed in **Recognition**, **Growth Opportunities**, **Accountability**, and **Overall Job Satisfaction**.

These findings suggest that departments with stronger employee engagement are characterized by greater employee recognition, clearer accountability, improved development opportunities, and higher overall job satisfaction.

Recommendations

Based on the analysis, the following recommendations are proposed:

- Strengthen employee recognition programs to acknowledge outstanding performance and improve morale.
- Expand professional development and career growth opportunities across all departments.
- Review leadership and performance management practices within lower performing departments.
- Encourage regular employee feedback to monitor engagement and identify emerging workplace concerns.
- Promote knowledge sharing by adopting best practices from the highest performing departments.
- Continue monitoring employee engagement through periodic surveys to measure the effectiveness of improvement initiatives.

Conclusion

This project successfully transformed raw employee survey responses into meaningful insights using Microsoft Power BI. Through data cleaning, modeling, DAX calculations, and dashboard development, the analysis provided a comprehensive overview of employee engagement across 21 departments.

The findings indicate that employees generally have a positive perception of their work environment, with a 77% positive response rate. Strong performance was observed in areas such as work expectations, supervisor support, and organizational purpose. However, opportunities exist to improve employee recognition, growth opportunities, accountability, and overall job satisfaction, particularly within lower performing departments.

The interactive dashboard enables management to monitor employee engagement, compare departmental performance, and identify key areas requiring improvement. By implementing the recommendations outlined in this report, the organization can strengthen employee satisfaction, enhance workplace culture, and support continuous organizational improvement.

Capstone Project Questions and Answers

Question 1

Which survey questions received the highest and lowest levels of agreement?

Answer

The analysis showed that employees expressed the highest levels of agreement with questions relating to:

- Work Expectations
- Supervisor Support
- Mission and Purpose
- Inclusion and Workforce Support

These results indicate that employees generally understand their responsibilities, feel supported by their supervisors, and recognize the importance of their work within the organization.

The lowest levels of agreement were observed for:

- Recognition and Praise
- Best Friend at Work
- Growth Opportunities
- Overall Job Satisfaction

These findings suggest opportunities for management to strengthen employee recognition programs, career development initiatives, workplace relationships, and employee engagement.

Question 2

Which departments performed the best and worst based on employee responses?

Answer

The dashboard identified the following departments as having the highest positive response rates:

Department	Positive Response Rate
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Communications Office	92%
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Emergency Management	90%
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Family Justice Center	89%
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Human Resources	89%
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Executive Office & Directors	89%
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These departments consistently demonstrated strong employee engagement and may serve as benchmarks for organizational best practices.

Departments with the highest negative response rates included:

Department	Negative Response Rate
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Sheriff's Department	49%
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District Court	26%
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Clerk of Superior Court	22%
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Planning and Public Works	21%
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Parks and Recreation	21%
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These departments present opportunities for targeted interventions to improve employee satisfaction and workplace culture.

Question 3

What insights can management gain from comparing the highest and lowest performing departments?

Answer

The comparison between the top five and bottom five departments revealed clear differences in employee perceptions across several survey questions.

The largest performance gaps were observed in:

- Recognition
- Growth Opportunities
- Accountability
- Overall Job Satisfaction

Departments with higher employee engagement consistently reported stronger positive responses in these areas, suggesting that effective leadership, employee recognition, and opportunities for professional development contribute significantly to overall employee satisfaction.

Management can use these findings to identify best practices from high performing departments and implement targeted improvement strategies within lower performing departments.

Additional Insights

Beyond the project requirements, several additional insights were identified during the analysis:

- The organization recorded an overall **77% positive response rate**, indicating generally favorable employee sentiment.
- Employees demonstrated a strong understanding of work expectations, with **93% positive responses**, making it the highest rated survey dimension.
- Positive responses declined from **Work Expectations (93%)** to **Opportunity to Do Best (81%)** and **Growth Opportunities (78%)**, indicating potential opportunities to strengthen employee development initiatives.
- The comparison between the top and bottom performing departments showed that **Recognition, Growth Opportunities, and Accountability** are key drivers of departmental performance.

- The interactive dashboard enables management to monitor employee engagement trends and supports data driven decision making through dynamic filtering by department and survey question.